
EMPLOYMENT

- 2022-present **Lafayette College**, Assistant Professor
2020-2022 **Swarthmore College**, Visiting Assistant Professor
2019-2020 **Colby College**, Visiting Assistant Professor
2018-2019 **Grinnell College**, Visiting Assistant Professor

EDUCATION

- 2018 **Ph.D. Mathematics**, University of California, Santa Barbara
Thesis: *The Geometry and Topology of the Dual Braid Complex*
Advisor: Jon McCammond
2014 **M.A. Mathematics**, University of California, Santa Barbara
2012 **B.A. Mathematics with Honors**, Grinnell College

GRANTS, AWARDS AND HONORS

- 2026-present **NSF Grant** (PI), "LEAPS-MPS: Geometry and Topology of Dual Artin Groups" (\$250,000)
2024-present **AIM SQuaRE**, "Dual Right-Angled Artin Groups"
with Rachael Boyd, Jon McCammond, Rose Morris-Wright, Chloé Postel-Vinay, Mireille Soergel
2026 **Marquis Distinguished Teaching Award**, Lafayette College
2021-2022 **MAA Project NExT Fellow** (Gold '21)
2020 **President's Fund for Racial Justice**, Swarthmore College
Course development grant: "Mathematics and Social Justice"
2018 **Excellence in Teaching Award**, UCSB Graduate Student Association
2017 **Outstanding Teaching Assistant**, UCSB Department of Mathematics
2012-2018 **Graduate Fellowship**, National Physical Sciences Consortium

PUBLICATIONS AND PREPRINTS

(mentored undergraduate coauthors denoted with an asterisk*)

13. **The Geometry of Rectangular Multisets**
submitted, with J. McCammond
12. **Noncrossing Partitions from Hull Configurations**
submitted, with G. Root*
11. **Continuous Noncrossing Partitions and Weighted Circular Factorizations**
submitted, with J. McCammond
10. **Geometric Combinatorics of Polynomials II: Polynomials and Cell Structures**
Advances in Mathematics, 502-111140 (2026), with J. McCammond
9. **Noncrossing Partitions from Cones and Semicircles**
Discrete Mathematics (2026), with K. Fang*, Y. Jiang*, E. Lin*, L. Lindenmuth*, E. Pokras*, G. Root*
8. **Noncrossing Partition Lattices from Planar Configurations**
Discrete & Computational Geometry, 75-1 (2026), with S. Cohen*, A. Harsh*, S. Martin*
7. **The Polyhedral Tree Complex**
Combinatorial Theory, 2-3 (2022)
6. **Geometric Combinatorics of Polynomials I: The Case of a Single Polynomial**
Journal of Algebra, 607-B (2022), with J. McCammond

5. **Boundary Braids**
Algebraic & Geometric Topology, 20-7 (2020), with J. McCammond, S. Witzel
4. **Critical Points, Critical Values, and a Determinant Identity for Complex Polynomials**
Proceedings of the American Mathematical Society, 148-12 (2020), with J. McCammond
3. **Undesired Parking Spaces and Contractible Pieces of the Noncrossing Partition Link**
Electronic Journal of Combinatorics, 25-1 (2018), with J. McCammond
2. **A Decomposition of Parking Functions by Undesired Spaces**
Electronic Journal of Combinatorics, 23-3 (2016), with M. Bruce*, M. Hlavacek*, R. Kudo*, I. Nicolas*
1. **Hankel Transforms of Linear Combinations of Catalan Numbers**
Journal of Integer Sequences, 14-11.5.1 (2011), with C. French, W. Qian, B. Saderholm

UNDERGRADUATE RESEARCHERS SUPERVISED

Lafayette College

Summer 2026 Mark McClair, Bitania Mesfin, Trung Nguyen, Rania Shuster

Spring 2025 Gina Root

Fall 2024 Kaiyi Fang, Yunting Jiang, Edgar Lin, Lucas Lindenmuth, Eleanor Pokras, Gina Root

Swarthmore College

Summer 2022 Stella Cohen, Andrew D. Harsh, and Spencer Park Martin

University of California, Santa Barbara

Summer 2015 Melody Bruce, Max Hlavacek, Ryo Kudo, and Ian Nicolas

TEACHING EXPERIENCE

Lafayette College

Math 104 A World of Mathematics	Spring 2023 (x2)
Math 162 Calculus II	Fall 2025 (x2), Spring 2024, Fall 2022
Math 165 Calculus I+ (Workshop Calculus)	Fall 2023
Math 263 Calculus III	Spring 2025 (x2), Fall 2024 (x2)
Math 325 Combinatorics	Fall 2025
Math 351 Abstract Algebra I	Fall 2023 (x2)
Math 352 Abstract Algebra II	Spring 2024
Math 358 Topology	Fall 2022
Math 378 Special Topic: Discrete and Computational Geometry	Fall 2024
Math 391 Independent Study (Lie Groups and Lie Algebras)	Fall 2023

Swarthmore College

Math 20 Mathematics and Social Justice	January 2021
Math 25 Single-Variable Calculus II	Fall 2021 (x2)
Math 27 Linear Algebra	Fall 2020
Math 67 Introduction to Modern Algebra	Spring 2022, Spring 2021 (x2)
Math 77 Advanced Topics in Algebra (Geometric Group Theory)	Spring 2022
Math 93 Directed Reading (Reflection Groups and Coxeter Groups)	Fall 2021

Colby College

Math 121	Single-Variable Calculus	Fall 2019 (x2)
Math 122	Series and Multi-Variable Calculus	Spring 2020 (x2)
Math 253	Linear Algebra	Spring 2020

Grinnell College

Math 131	Calculus I	Fall 2018 (x2)
Math 133	Calculus II	Spring 2019, Fall 2018
Math 444	Senior Seminar (Geometric Group Theory)	Spring 2019

University of California, Santa Barbara (Instructor of Record)

Math 4A	Linear Algebra with Applications	Summer 2017, Summer 2016
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University of California, Santa Barbara (Teaching Assistant)

Math 34A	Calculus for the Life and Social Sciences (Head TA)	Winter 2015
Math 3A	Calculus I	Winter 2016
Math 3B	Calculus II	Winter 2016
Math 4A	Linear Algebra with Applications	Spring 2017, Spring 2016
Math 6A	Vector Calculus I	Fall 2016
Math 8	Transition to Higher Mathematics	Spring 2018, Winter 2017, Spring 2015, Fall 2014, Winter 2014, Fall 2013
Math 108A	Advanced Linear Algebra I	Fall 2017, Fall 2015, Spring 2014

SELECTED RESEARCH TALKS

Invited Talks (Conferences and Colloquia)

2026	Early Career Complex Dynamics Workshop	University of Michigan
2026	Lehigh Math Colloquium	Lehigh University
2026	Joint Mathematics Meetings AMS Special Session on Algebraic and Geometric Combinatorics	Washington, DC
2024	AMS Fall Eastern Sectional Meeting Special Session on Geometric Group Theory	Albany, NY
2024	BIRS Workshop: Combinatorial Nonpositive Curvature	Banff, Canada
2023	Spring Topology and Dynamical Systems Conference Special Session on Geometric Group Theory	Rhodes College (virtual)
2022	Summer@ICERM REU	Providence, RI
2022	Bi-College Math Colloquium	Haverford College
2021	AMS Spring Southeastern Sectional Meeting Special Session on Groups, Geometry, and Topology	Atlanta, GA (virtual)
2014	Mathematics and Statistics Student Seminar	Grinnell College

Invited Talks (Research Seminars)

2026	University of Glasgow	Geometry & Topology Seminar
2025	The Ohio State University	Geometric Group Theory Seminar
2025	University of Iowa	Geometry & Topology Seminar
2025	Temple University	Geometry & Topology Seminar
2023	University of California, Santa Barbara	Topology Seminar
2023	University at Albany	Algebra/Topology Seminar
2022	Haverford College	PATCH - Philadelphia Area Topology (Contact/Hyperbolic)
2022	University of Pennsylvania	Deformation Theory Seminar (virtual)
2022	Temple University	Geometry & Topology Seminar
2020	Otto-von-Guericke-Universität Magdeburg	Algebra & Geometry Seminar (virtual)
2019	Georgia Institute of Technology	Geometry & Topology Seminar
2018	Brown University	Geometry & Topology Seminar

Contributed Conference Talks

2020	Joint Mathematics Meetings AMS Special Session on The Geometry of Complex Polynomials and Rational Functions	Denver, CO
2019	Tech Topology Conference	Atlanta, GA
2019	Algorithms in Complex Dynamics and Mapping Class Groups	Providence, RI
2018	Tech Topology Conference	Atlanta, GA
2017	Tech Topology Conference	Atlanta, GA
2016	Tech Topology Conference	Atlanta, GA
2016	Topology Students Workshop	Atlanta, GA

DEPARTMENT SERVICE

Seminar and Colloquium Organization

2023-present	Colloquium Co-Organizer	Lafayette College
2020-2022	Colloquium Helper	Swarthmore College
2018-2019	Colloquium Organizer (<i>Mathematics and Statistics Student Seminar</i>)	Grinnell College
2013-2018	Seminar Organizer (<i>Discrete Geometry & Combinatorics Seminar</i>)	UC Santa Barbara
2012-2017	Seminar Co-Organizer (<i>Hypatian Seminar</i>)	UC Santa Barbara

Committee Service

2024-2025	Tenure-Track Search Committee	Lafayette College
2023-2024	Linear Algebra Committee	Lafayette College

Events and Community Building

2023-2024	Teaching Lunch Co-Organizer	Lafayette College
2023-2024	Problem Group Co-Organizer	Lafayette College
2020-2022	Community Building Helper	Swarthmore College

Department Talks Given

2025	Mathematical Adventures and Diversions: Introduction to Noncrossing Partitions	Lafayette College
2023	Pi Mu Epsilon Induction Ceremony	Lafayette College
2023	Cryptography Competition Kickoff Talk: Diffie-Hellman Key Exchange	Lafayette College
2022	Mathematical Adventures and Diversions: Configuration Spaces	Lafayette College
2021	Interactive Math Colloquium	Swarthmore College
2019	New Math Faculty Colloquium	Colby College

Other Contributions

2023-present	Teaching Repository Manager	Lafayette College
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COLLEGE SERVICE

Lafayette Undergraduate Thesis Committees

2025	Hanzhe Zhou (Computer Science and Mathematical Sciences, co-advisor) <i>Convex Flip Distance and Representations of Triangulations</i>
2023	Brian Payne (Music) <i>A Blue Twilight</i>

Committee Service

2025	Strategic Plan Implementation Action Team <i>Democracy and its Technologies: Pedagogy and Curriculum</i>
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Interdisciplinary Activities

2024	Co-Organizer , Math and Art Workshop with guest artist Gabriel Rico, Lafayette College
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PROFESSIONAL SERVICE

Conference Organization

2022	Session Co-Organizer (<i>Teaching Undergraduate Topology</i>), Nearly Carbon Neutral Geometric Topology Conference
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Peer Review

Journal Referee (*Journal of Combinatorial Algebra*, *Journal of Combinatorics*, *Proceedings of the AMS*, *Rocky Mountain Journal of Mathematics*)
Reviewer (six articles and one book in *Mathematical Reviews*)

PEDAGOGICAL DEVELOPMENT AND OUTREACH

2023-2025	Invited Project Participant , <i>Framing Mathematics as a Foundation for Ethical STEM</i>
Winter 2018	Mathlab Coordinator , UC Santa Barbara
2016-2017	Lead Teaching Assistant (<i>Math Department TA Training</i>), UC Santa Barbara
2016	Session Speaker (<i>Campus-wide orientation for new teaching assistants</i>), UC Santa Barbara
2013-2015	Math Circle Instructor (4 talks given), UC Santa Barbara

CONFERENCES AND WORKSHOPS ATTENDED

2026	Early Career Complex Dynamics Workshop	Ann Arbor, MI
2026	Joint Mathematics Meetings	Washington, DC
2024	AMS Fall Eastern Sectional Meeting	Albany, NY
2024	Banff International Research Station: Combinatorial Nonpositive Curvature	Banff, Canada
2024	Hot Topics: Artin Groups and Arrangements - Topology, Geometry, and Combinatorics	Berkeley, CA
2023	AIM Workshop - Geometry and Topology of Artin Groups	Pasadena, CA
2023	Framing Mathematics as a Foundation for Ethical STEM	Washington, DC
2023	Spring Topology and Dynamical Systems Conference	(virtual)
2022	Nearly Carbon Neutral Geometric Topology Conference	(virtual)
2022	MAA MathFest	Philadelphia, PA
2021	MAA MathFest	(virtual)
2021	Artin Groups, CAT(0) Geometry, and Related Topics	Columbus, OH (virtual)
2021	Perspectives on Artin Groups	Edinburgh, Scotland (virtual)
2021	AMS Spring Southeastern Sectional Meeting	Atlanta, GA (virtual)
2020	Joint Mathematics Meetings	Denver, CO
2019	Tech Topology Conference	Atlanta, GA
2019	ICERM Workshop - Algorithms in Complex Dynamics and Mapping Class Groups	Providence, RI
2018	Tech Topology Conference	Atlanta, GA
2018	Young Geometric Group Theory VII	Les Diablerets, Switzerland
2017	Tech Topology Conference	Atlanta, GA
2017	Algebraic and Geometric Combinatorics of Reflection Groups	Montréal, Canada
2017	Young Geometric Group Theory VI	Oxford, England
2017	Joint Mathematics Meetings	Atlanta, GA
2016	Tech Topology Conference	Atlanta, GA
2016	MSRI Workshop - Geometry of Mapping Class Groups and $\text{Out}(\mathbb{F}_n)$	Berkeley, CA
2016	FPSAC (Formal Power Series and Algebraic Combinatorics)	Vancouver, Canada
2016	Topology Students Workshop	Atlanta, GA
2015	AMS Spring Western Sectional Meeting	Las Vegas, NV
2014	The Thin Manifold - Graduate Workshop	Iowa City, IA
2014	Joint Mathematics Meetings	Baltimore, MD
2013	A Day of Triangulations	Los Angeles, CA
2013	Joint Mathematics Meetings	San Diego, CA
2010	Iowa Summer Research Symposium	Ames, IA